Effects of memory for discourse context on syntactic parsing

Adults use referential and discourse context, along with new information, to aid sentence processing, whereas children prefer syntactic structure and global knowledge [1-2, 6]. Children's preference for bottom-up cues is attributed to cue reliability, memory shortcomings, and processing speed [2, 3-5]. However, when learning certain words or parsing specific sentences, children *do* integrate top-down cues [7-11]. Speech perception suggests children utilize top-down information when bottom-up cues are unreliable (McGurk effect; Ganong effect) [12-13]. It remains unclear how though, and in what contexts, children use top-down cues. To investigate this question, we focused on memory and examined children's word-learning in English in two linguistic contexts, one globally ambiguous and one temporarily so. It could be that children do not recruit memories of prior discourse during parsing, but they *might* in ambiguous contexts.

Children (n=51, 4;0 to 7;5) completed two experiments, across two presentations, receiving different conditions between the presentations. Time between Presentation 1 and 2 ranged from 1 to 86 days (mean=14.87). In Experiment 1, globally ambiguous, children heard sentences like (1) while eye-movements and actions were measured in response to independently-normed objects (Fig. 1): 1) <u>modifier</u>: an unfamiliar object alongside a known animal, 2) <u>instrument</u>: an unfamiliar object featuring tool-like properties. 25 children received the instrument condition in the first presentation and modifier in the second; 26 received the conditions in opposite order. In Experiment 2, temporarily ambiguous, children heard sentences like (2). Responses were measured to independently-normed creatures (Fig. 2): 1) patient, relatively small, and 2) agent, relatively large. Sentences tested agent-first bias, varying novel word position and voice. 29 children received the novel word first condition in the first presentation and the novel word second condition in the second; 22 received the conditions in opposite order.

For E1, instrument responses were coded as -1 and modifier responses as 1. Instrument preference was as the mean response per presentation. For E2, agent responses were coded as -1 and patient responses as 1. Agent-first bias was measured depending on condition. In the NP1 condition, it was mean response for active trials minus mean response for passive trials; in NP2, it was mean response for passive trials minus mean response for active trials. Eye-tracking analyses are ongoing.

In Presentation 1 of E1, children hearing instrument verbs primarily selected instruments, and children hearing modifier verbs modifiers (Fig. 3). But, during Presentation 2, children hearing modifier verbs usually selected *instruments*. This verb x presentation interaction ($p<0.01$) suggests children prefer scene memory to syntactic structure in this globally ambiguous context. In Presentation 1 of E2, children in the NP1 condition generally select agents during active trials and patients during passive trials, while children in the NP2 condition select agents during passive trials and patients during passive trials (Fig. 4). This trend continues in Presentation 2: there is no novel word position x presentation interaction ($p=0.801$). Figures 5 and 6 show magnitude of change in instrument preference and agent-first preference, respectively, against days between Presentations 1 and 2. There is no relationship ($p=0.48$; $p=0.76$), suggesting changes in instrument response during E1 are not attributable to syntactic priming. Children utilize syntactic structure to parse sentences in a temporarily ambiguous context, but rely on scene memory in a globally ambiguous context. Altogether, results indicate children possess more sophisticated discourse-integration abilities than previously suggested and a need to incorporate discourse-integration *and* generalized syntactic structure into child language development theory.

1. Snedeker, J., & Trueswell, J. C. (2004). *Cognitive Psychology, 49*(3). **2.** Yacovone, A., Shafto, C. L., Worek, A., & Snedeker, J. (2021). *Cognitive Psychology, 131*. **3.** Gleitman, L. (1990). *Language Acquisition, 1*(1). **4.** Gertner, Y., & Fisher, C. (2012). *Cognition, 124*(1). **5.** Kam, C. L. H., & Newport, E. (2005). *Language Learning and Development, 1*(2). **6.** Yazbec, A. Kaschak, M. P., & Borovsky, A. (2019). *Cognitive Science, 43*(1). **7.** Hacquard, V. (2023). *Journal of Child Language, 50*(5). **8.** Author & Snedeker, J. (Date). *Developmental Psychology, 49*(6). **9.** Author & Ovans, Z. (Date). *Cognitive Science, 46*(1). **10.** Yu, Y., Havron, N., & Fisher, C. (2025). *Language Learning, 75*(1). **11.** Fisher, C., Jin, K., & Scott, R. M. (2020). *Topics in Cognitive Science, 12*(1). **12.** McGurk, H. & MacDonald, J. (1976). *Nature*. **13.** Ganong, W. (1980). *J. of Exp. Psych. (6)*.

1a. Lena is going to hit the bear with the avthut.
1b. Lena is going to find the bear with the ekrop.

2. Experiment 2 sentences

	NP1 (novel noun first)	**NP2 (known noun first)**
Active	The blicket will be quickly eating the seal.	The seal will be quickly eating the tharklin.
Passive	The bellwer will be quickly chased by the fox.	The fox will be quickly chased by the akburl.

Figure 1: Experiment 1 display

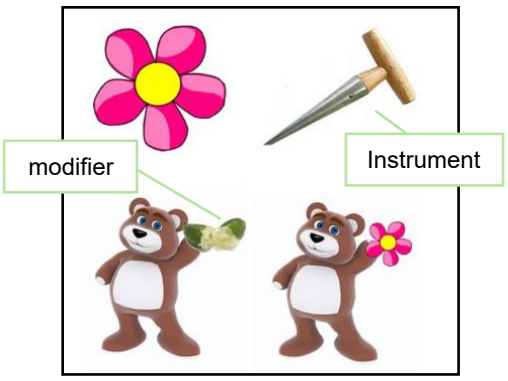


Figure 2: Experiment 2 display

Figure 3

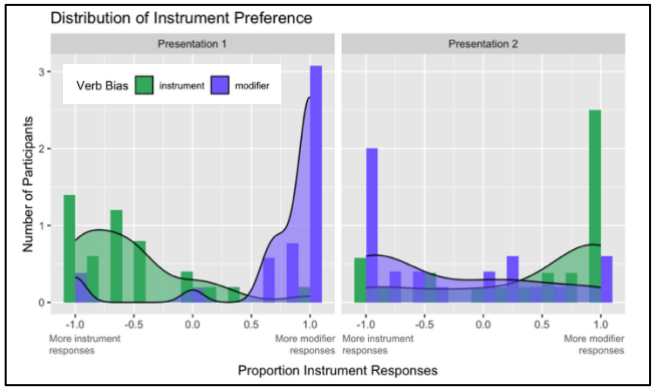
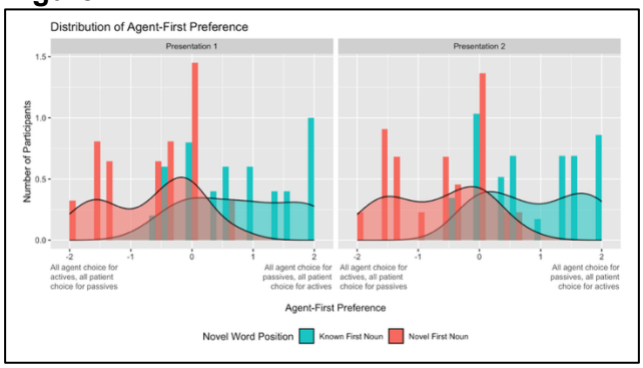
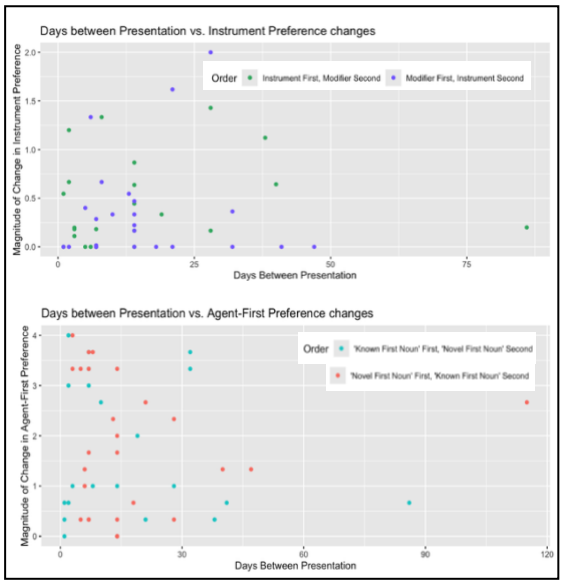


Figure 4



Figures 5 and 6